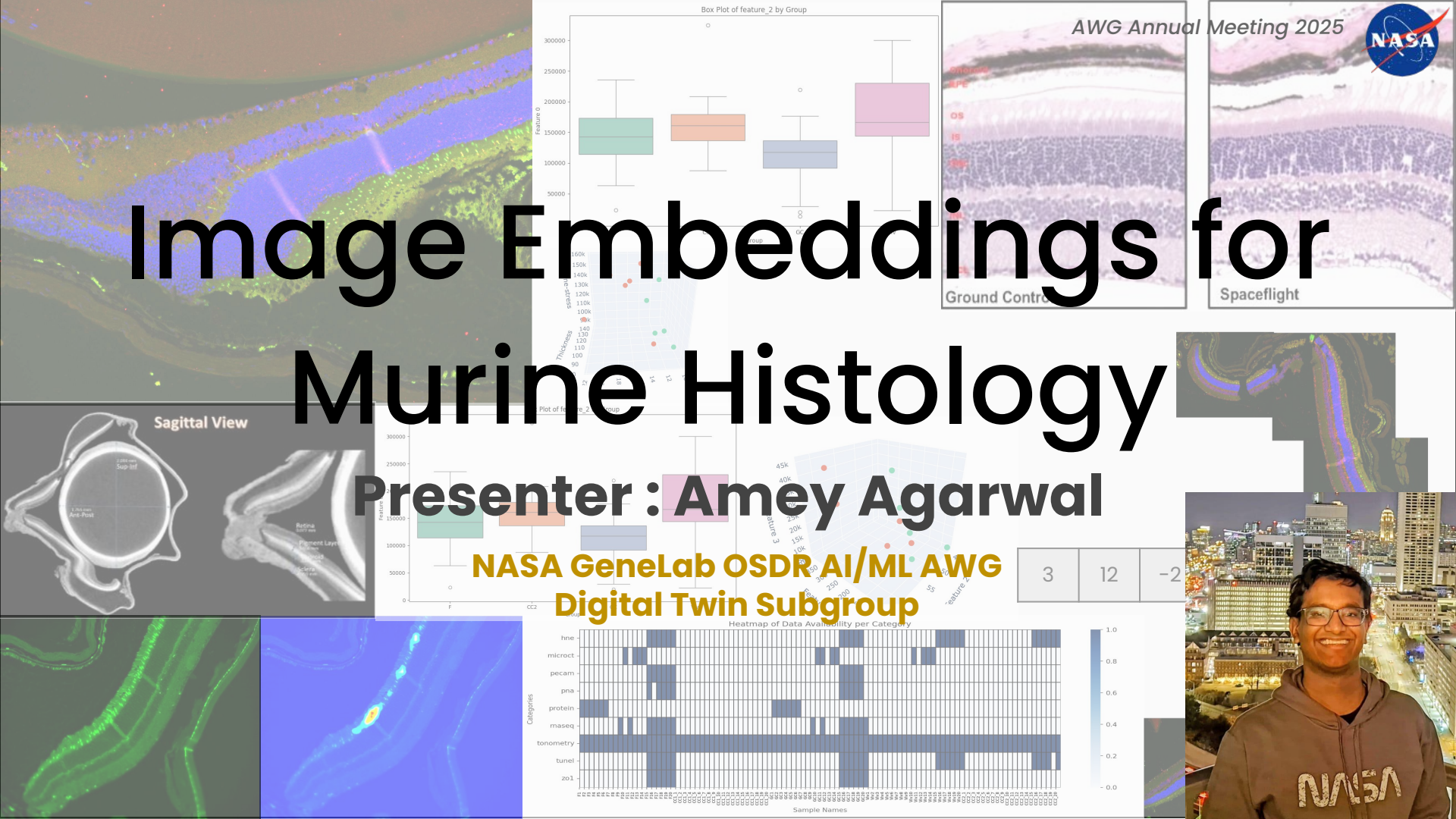
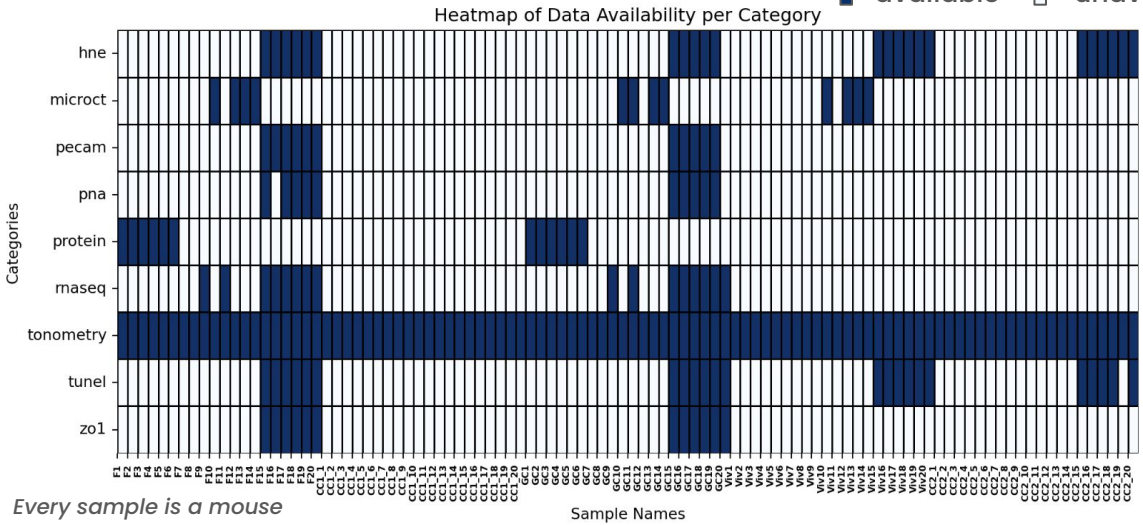


Image Embeddings for Murine Histology

Presenter : Amey Agarwal

NASA GeneLab OSDR AI/ML AWG
Digital Twin Subgroup

Image Embeddings Project



Every sample is a mouse

AI/ML : Digital Twin Subgroup

Imputation

Data Sparsity

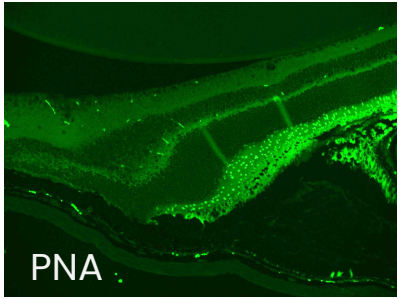
Better Data
Better Imputation

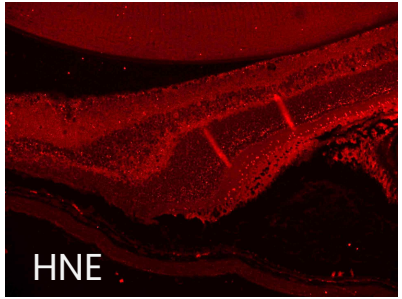
Re derive given numerical data from raw img

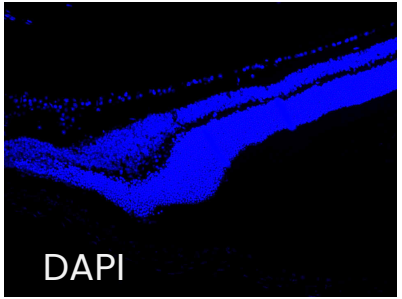
OSD-557 Version 3

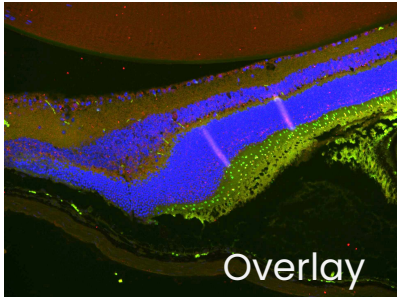
Spaceflight influences gene expression, photoreceptor integrity, and oxidative stress related damage in the murine retina (RR-9)

Sample F15/
› Sample-file-1/
› Sample-file-n/









183 Sample-file folder total

The main idea

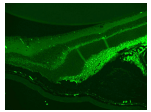


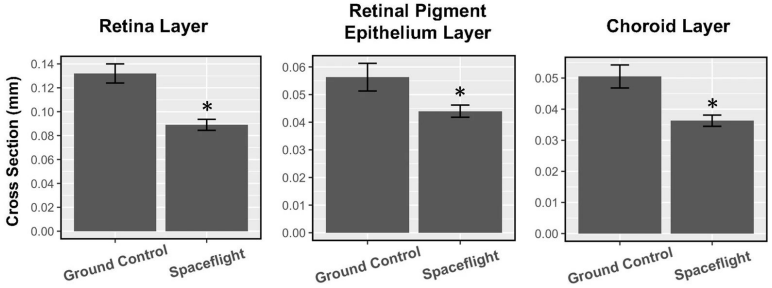
Image (1440, 1920)

3	12	4	15	-2	

Embedding (1 , 1024)

How do we know we have good embeddings? - Statistically valid

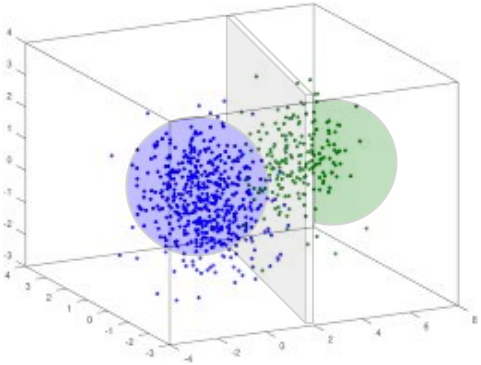
Published Paper Findings on Data



Eliah G. Overbey *et al. "Spaceflight influences gene expression, photoreceptor integrity, and oxidative stress-related damage in the murine retina"*

F and GC have a distinction

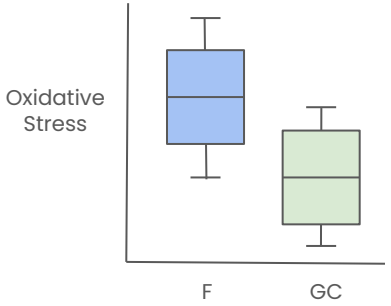
1) Decision Boundary



F

GC

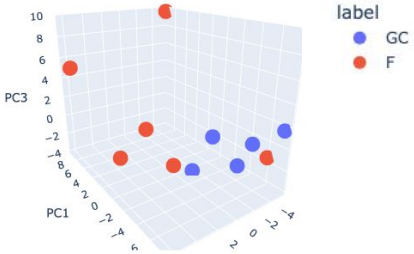
2)



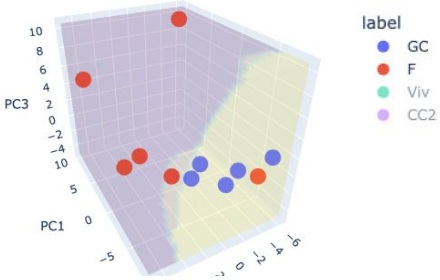
Intuitively we start with picking area and intensity as features to encode

A - Area + Intensity

A



Decision Boundary



SVM Accuracy : 75%

Embedding

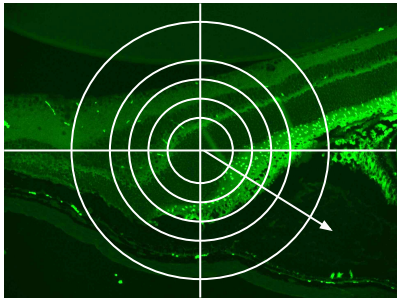
3	12	-2	

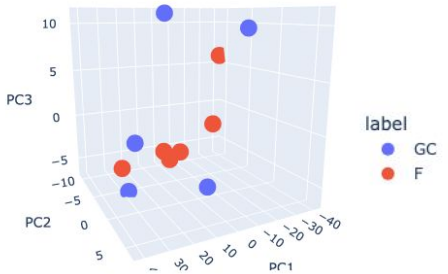
Plot ↓ PCA

X	Y	Z	

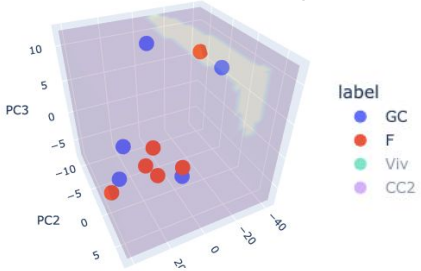
Radial Intensity Histogram

B





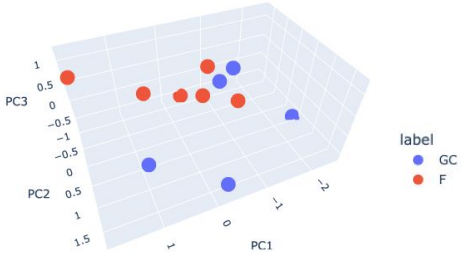
Decision Boundary



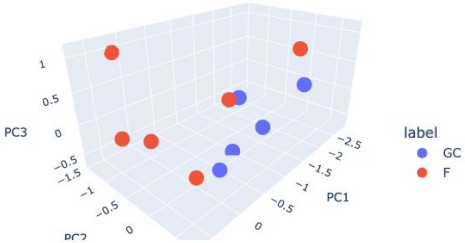
SVM Accuracy : 50%
(no proper decision boundary)

Thickness Cell Density Intensity

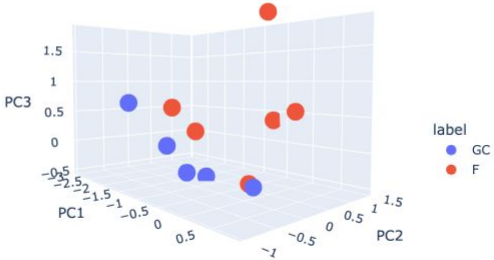
C



v1



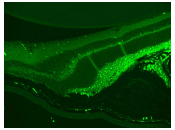
v2

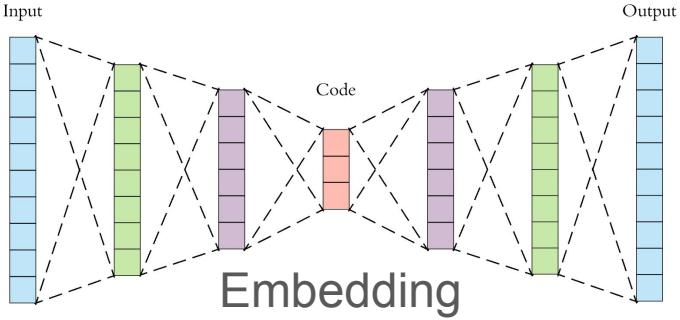


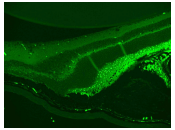
v3

AUTO-ENCODERS

D

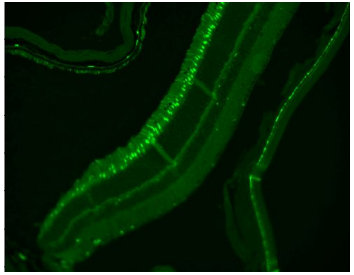




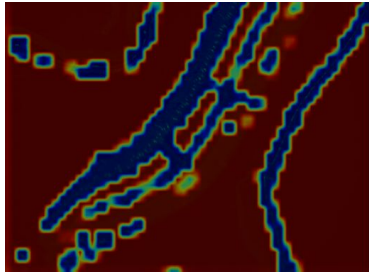


D

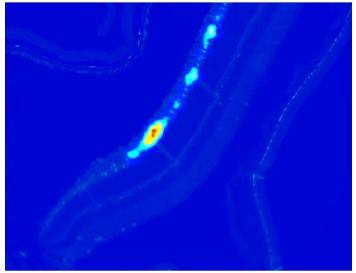
Original Image



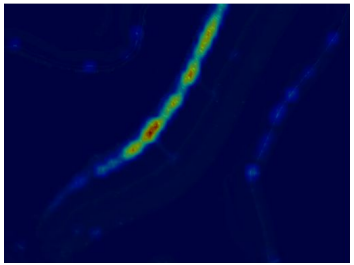
GradCAM(Embedding=0)



GradCAM(Embedding=805)

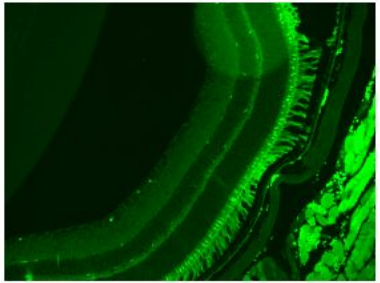


GradCAM(Embedding=1023)

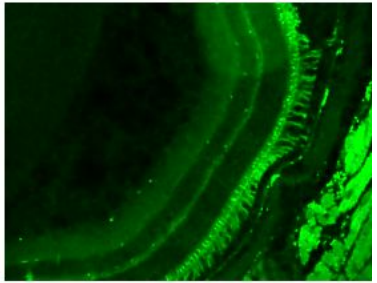


What if I change an embedding and then decode the image?

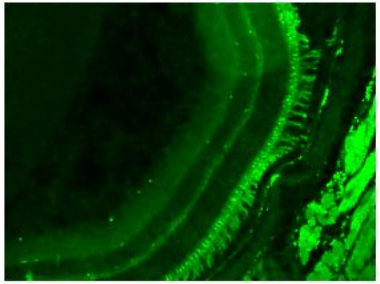
F



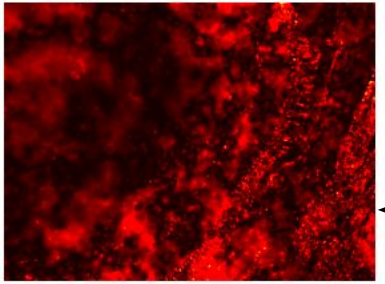
Recons Acc: 98.60% wrt col1

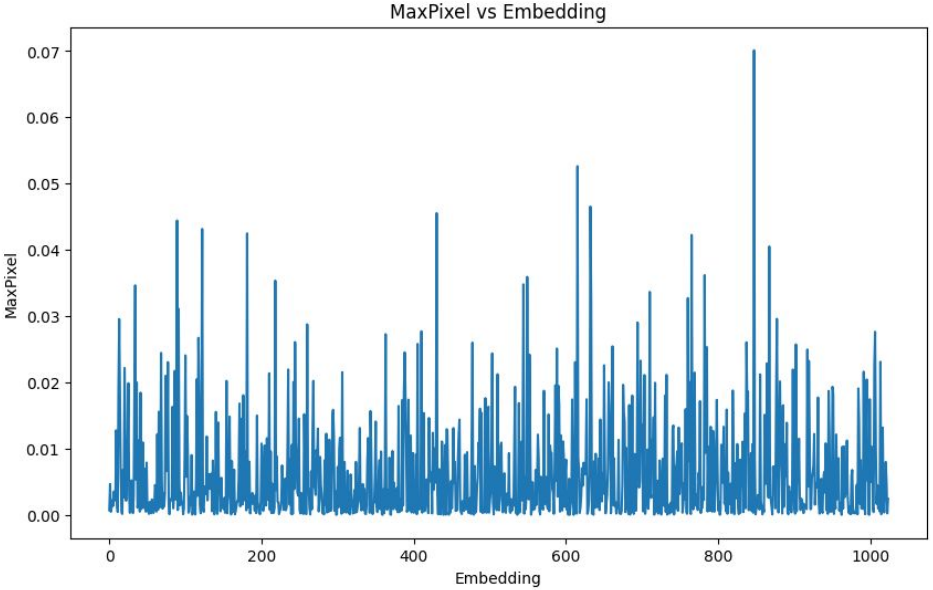


Recons Acc: 100.00% wrt col2



Embding: 0; maxPixelvalue:0.0042





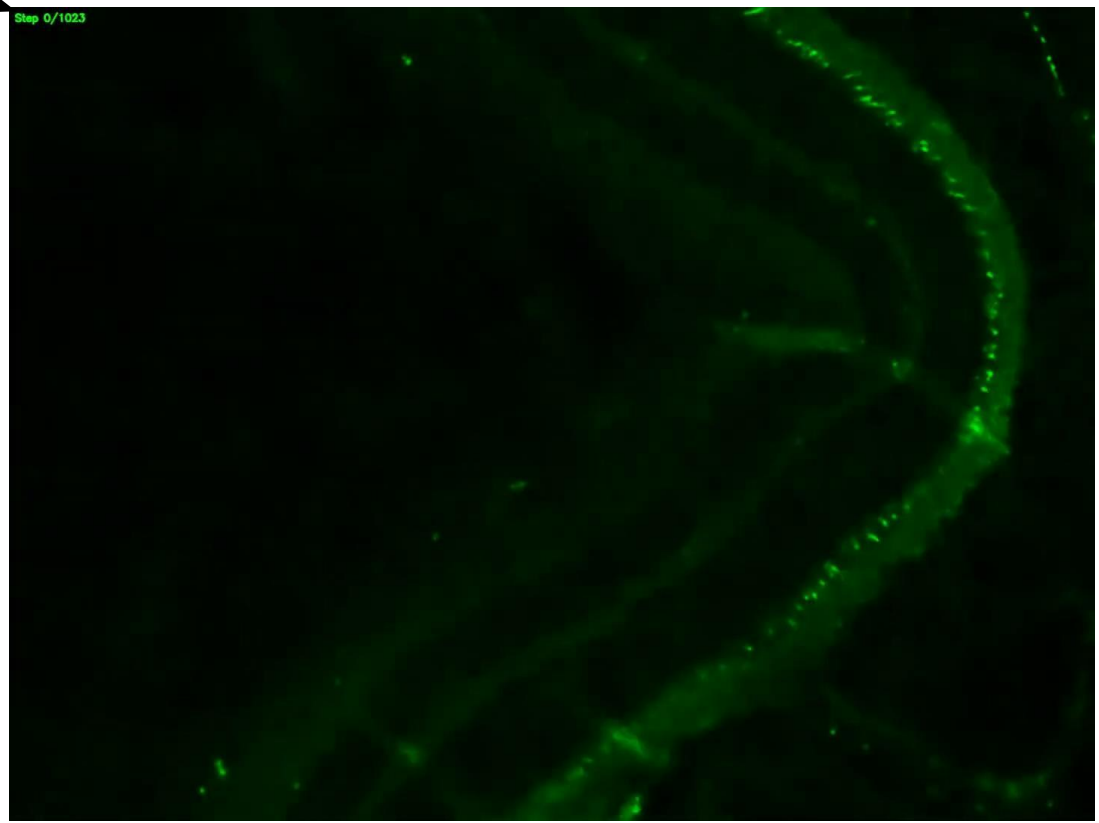
Pixel Intensity Change vs Embedding Change

Autoencoder Embeddings Video

Increase Screen Brightness
Open video in full screen
The changes in image come very gradually

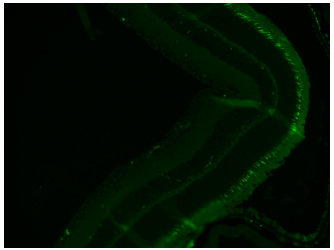
Step Count

Stepwise transition of decoded embeddings from a GC retinal image to a F retinal image

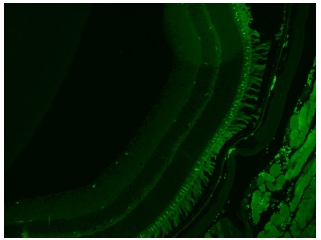


QR Code to video

Mapping from Ground control to Flight



GC



F

Why?
Visualising the intermediate changes in the embeddings to understand what the autoencoder has learnt

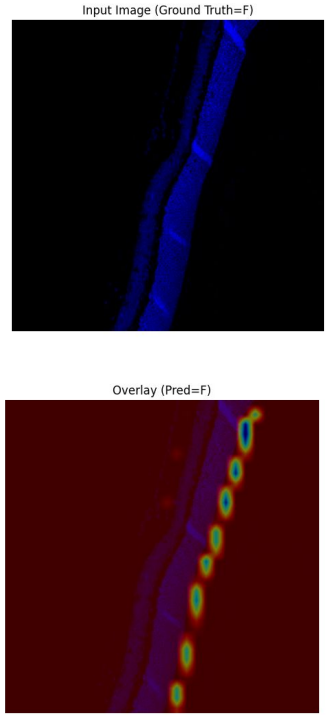
Takeaways

- Embeddings are impactful with the following:
  - Represent an affine digital twin that can detect retinal anomaly
  - numerical method provides for ease of use with ML techniques
  - the embedding arrays can act as pseudo-measures to provide a scoring metric to eye structure and changes

Impact & Limitations

- Image embedding is a **convenient technique** in representing image features succinctly
- **Expert guided-based** feature extraction techniques are very helpful in focusing on the important features in the space biology dataset
- **Explainability** is an imperative concept to consider the embeddings for practical use
- A few **factors** in space biology datasets that may **hamper** intended comparisons or results include:
  - aging of the biological samples
  - outliers

D



Thank you.

Big thanks to Ryan Scott and my Subgroup leads and members - Jian Gong, Lauren Sanders, Vaishnavi Nagesh, Pinar, Sylvain Costes and the entire Digital Twin Subgroup !

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